

Kitchcu

Product Depth Complete Guide

Modules | Features | Flows | Architecture | APIs | Build Status | Roadmap | KPIs

Kitchcu cloud kitchen platform

Audience: CPO, Product, Engineering, Investors, Partners

- 48-feature catalog with implementation status (Sprints 1-4)
- Owner and customer pain points mapped to modules and code
- End-to-end application and event-driven data flows
- Full API reference, database schemas, and service topology
- Business model, KPIs, competitive position, and roadmap

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7. Part VII - Business, Delivery & Metrics (Model, Roadmap, KPIs)

Product Foundation

Kitchcu is a B2B2C operating system for cloud kitchens - not a food aggregator. Owners capture WhatsApp and manual orders, manage honest menus with live-capture photos, and track order lifecycle. Phase 2 adds customer discovery, CRM, and analytics.

WhatsApp to reveal

Owner promise

Live photos + fair

Customer promise

Subscription, 0%

Platform model

10 pilot kitchens / 3 mo

MVP target

Differentiation

- WhatsApp-native operations (70%+ of owner orders today)
- Live-capture hero images enforced in catalog API
- Owner-owned CRM and customer data (Phase 2)
- Zero food commission - subscription SaaS model

2. Personas & Principles

Persona	Goal	Surface Today	Planned
Raj (Owner)	Run kitchen	kitchen.kitchcu.in PWA	Reports LIVE
Priya (Customer)	Trusted food	customer.kitchcu.in PWA	Browse LIVE; checkout P2
Admin	Moderate/support	admin.kitchcu.in	Tickets LIVE

Non-Negotiable Principles

- Truth in media - hero photos must be live-capture
- Quality over speed - owner-set prep times
- Owner sovereignty - CRM belongs to kitchen
- Progressive complexity - unlock after traction
- WhatsApp-native - primary intake channel

PART 2

Market Problems & Solutions

ID	Pain	Module	Solution	Status
P1	WhatsApp chaos	Order+Notify	Parser, drafts	Partial
P2	Commission 18-30%	Billing	Subscription	S6
P3	No profit view	Analytics	Owner revenue LIVE	Partial
P4	Stock photos	Catalog	Live-capture	Done
P5	Taste variance	Catalog	Quality fields	Partial
P6	No CRM	Marketing	Owner CRM	P2
P7	Guesswork promos	Growth	Suggestions	P2
P8	Channel silos	Order	One lifecycle	Partial

4. Customer Pain Points (C1-C6)

ID	Pain	Answer	Status
C1	Untrustworthy photos	Live-capture media	Done
C2	Opaque fees	PostGIS + owner rules	Fields only
C3	No tracking	Lifecycle + notify	Partial
C4	Generic ratings	Home-taste 1-5	P2
C5	One kitchen/cart	Multi-kitchen checkout	P2
C6	No tiffin	Meal plans	P2

Platform Modules

Gateway :18000 -> Identity :18001 | Catalog :18002 | Order :18003 | Notification :18005
PostgreSQL 16 + PostGIS (schema-per-domain) | Redis Streams + cache | MinIO media

Module	Schema	Status	Events
Identity	ckac_identity	DONE S1	kitchen.created
Catalog	ckac_catalog	DONE S2	dish.created/updated
Order	ckac_orders	DONE S3	order.placed, status
Notification	ckac_support	DONE S4	whatsapp + support.ticket.*
Billing	ckac_billing	S6	payment.captured
Analytics	order svc	S5 PARTIAL	owner revenue/segments
Marketing	ckac_marketing	P2	CRM, coupons

6. Service Deep Dives

Identity - Owners, Kitchens, OTP, JWT

- Kitchen code: CKPNQ001 format; PostGIS location on create
- APIs: register, OTP, POST/GET kitchens, owners/me

Catalog - Menu, Categories, Live Media (F13-F15)

- 10 default categories auto-seeded; menu Redis cache TTL 300s
- Hero image validator rejects non-live-capture; PATCH dish updates

Order - Intake, Lifecycle, History (F03-F05)

- Manual orders, message parser, drafts, confirm flow
- State machine: received -> ... -> delivered | cancelled
- Order code: {kitchen_code}-BILL-{date}-{seq}

Notification - WhatsApp, AI Support Chat, Tickets (F01, F45)

- Meta webhook; kitchen lookup by whatsapp_phone_id
- POST /support/chat - owner & customer modes with AI escalation
- POST /support/tickets - public create; admin CRUD via /admin/tickets
- Schema ckac_support: support_tickets, support_ticket_messages

48-Feature Catalog

ID	Feature	Status
F01	WhatsApp capture	PARTIAL
F02	Message parser	DONE
F03	Manual order	DONE
F04	Lifecycle	DONE
F05	Order history	DONE
F07	Revenue report	PARTIAL
F13-F15	Menu + live photo + categories	DONE
F26	Subscription	PARTIAL
F30	Prep time	DONE
F42-F43	Payments	S6
F45	Notify + AI chat + tickets	PARTIAL

8

Done

4

Partial

48 features

Total

8. Phase 2 & 3 Features

Phase 2 - Growth

- F06 Multi-kitchen checkout | F16-F18 Home taste ratings
- F27-F33 Delivery + customer PWA | F34-F40 CRM, tiffin, coupons
- F44 Split payment via Razorpay Route

Phase 3 - Differentiation

- F19 Ingredient mapper | F21-F24 Learning + chef rankings
- F46-F48 Live kitchen streaming premium

PART 5

Application & Data Flows

WhatsApp Order Intake

```
Webhook -> notification -> whatsapp.message.received  
-> order internal API -> draft -> owner confirm -> order.placed
```

Order Lifecycle

```
PATCH status -> validate transition -> order_status_events  
-> order.status.changed -> (future) customer notify
```

Event Catalog (EDD + Outbox)

Event	Producer	Stream
kitchen.created	identity	ckac:identity:kitchen
dish.created/updated	catalog	ckac:catalog:dish
order.placed	order	ckac:orders:order
order.status.changed	order	ckac:orders:order
order.draft.created	order	ckac:orders:draft
whatsapp.message.received	notification	ckac:notify:whatsapp

Technical Reference

Schema	Tables	Status
ckac_identity	owners, kitchens	LIVE
ckac_catalog	categories, dishes, dish_media	LIVE
ckac_orders	orders, items, events, drafts	LIVE
ckac_events	outbox, processed_events	LIVE
ckac_support	support_tickets, messages	LIVE

API Summary (Gateway :18000/api/v1)

Identity: /auth/* /owners/* /kitchens
 Catalog: /kitchens/{id}/categories|menu|dishes
 Order: /kitchens/{id}/orders/* /orders/{id}/status /analytics/*
 Notification: /webhooks/whatsapp /support/chat /support/tickets
 Admin: /admin/tickets/*

11. Technology Stack

Layer	Choice
Backend	Python 3.12 FastAPI microservices
Database	PostgreSQL 16 + PostGIS
Events/Cache	Redis Streams + menu cache
Frontend	React + Vite + TS PWAs (portal, custom
Payments	Razorpay Route (S6)
Infra	Docker Compose -> Kubernetes

Business & Metrics

Tier	Price/mo	Includes
Starter	Rs 499	WhatsApp, menu, basic reports
Pro	Rs 1,499	CRM, marketing, analytics
Enterprise	Rs 3,999	API, multi-branch, stream

Zero food commission. Kitchen keeps 100% of food revenue.

13. Build Status & Roadmap

S1-S4 done; S5 partial

Sprints done

90+ passing

Tests

5 + gateway

Services

Sprint	Deliverable	Status
S1-S4	Identity, Catalog, Order, Notification	COMPLETE
S5	PWAs + owner analytics + support	PARTIAL
S6	Billing + revenue	Planned

14. KPIs & Competitive Position

Metric	Target
Time to first order	< 5 min
WhatsApp parse rate	95%
Hero live-capture	100%
Owner retention M6	80%

vs Aggregators: zero commission, owner CRM, live-capture. vs POS: customer discovery + trust layer + WhatsApp analytics.

Kitchcu

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See CKAC-PRODUCT-DEPTH-GUIDE.md for markdown source.